

Notes: Hoffman, Kahn, and Li (QJE, 2018)

Discretion in Hiring

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1 Big picture

- **Question.** How should firms combine algorithmic job-test recommendations with human judgment in hiring? The paper asks whether managers improve hiring outcomes when they overrule test scores, or whether discretion mainly introduces bias and mistakes.
- **Setting.** The authors study the introduction of a job test across 15 client firms employing low-skilled service-sector workers in a fairly standardized job. The test gives each applicant a green, yellow, or red recommendation.
- **Core empirical setting.** Managers are shown the test score and encouraged to use it, but they are not forced to hire strictly by the test. This creates variation in whether managers follow or overrule algorithmic recommendations.
- **Main test-adoption result.** Using staggered adoption of job testing across locations, the paper finds that workers hired after testing is introduced stay substantially longer. With full controls, the estimated effect is about 0.23 log points, or a little over 25% longer tenure.
- **Main discretion result.** Managers who make more exceptions to the test recommendations hire workers who leave more quickly. A one-standard-deviation increase in manager exception rates is associated with roughly 6% shorter job durations in the full post-testing specification.
- **Mechanism.** The paper interprets this negative exception-outcome relationship as evidence that managers often overrule the test not because they have superior private information, but because they are biased or mistaken.
- **Robustness.** Passed-over higher-scoring applicants later perform better than lower-scoring exception hires. The paper also finds no evidence that high-exception managers sacrifice tenure in exchange for higher output per hour.
- **Economic takeaway.** In this standardized low-skilled service setting, job testing improves hiring and human discretion often weakens the value of the test. The firm may improve outcomes by limiting discretion, especially for managers who frequently hire against test recommendations.

2 Incremental contribution of the paper

- **From “algorithms help” to “how should humans use algorithms?”** Prior work shows that screening tests can predict worker outcomes. This paper goes one step further by asking whether managers add value when they are allowed to override an algorithmic recommendation.
- **A direct study of discretion in hiring.** The paper provides rare empirical evidence on the allocation of decision rights in hiring: should the firm rely on a test, or allow local HR managers to use subjective judgment?
- **Unique applicant-level data.** The data include applicants, hires, test-score categories, managers, locations, and worker outcomes. This lets the authors compare not only hired workers, but also the pools from which managers choose.
- **New exception-rate measure.** The paper constructs an empirical measure of hiring against the test: hiring a lower-score applicant while a higher-score applicant from the same pool is not hired.

- **Model-based interpretation.** The theoretical model clarifies why exception rates are informative. If exceptions reflect valuable private information, higher-exception managers should have better hires. If exceptions reflect bias or mistakes, higher-exception managers should have worse hires.
- **Connection to organizational economics.** The paper turns a classic authority/delegation problem into a measurable empirical question: how much discretion should a principal give to agents who may have private information but also misaligned preferences or flawed judgment?
- **Contribution to workforce analytics.** As firms increasingly adopt tests, algorithms, and analytics, the paper shows that the technology’s value depends on the organizational process around it. A good test can be undermined if managers frequently ignore it.
- **Practical managerial contribution.** The results suggest that firms should monitor how managers use decision aids, provide feedback, and consider limiting discretion when exceptions systematically predict worse outcomes.

3 Data and measurement

3.1 Job test and testing adoption

- The data come from an anonymous job-testing provider, called the data firm in the paper.
- The provider sells a test for a low-skilled service-sector job. The job is similar to data entry, standardized test grading, or call-center work: workers perform a fairly standardized individual task.
- The test contains a large online questionnaire covering computer or technical skills, personality, cognitive skills, fit for the job, and job scenarios.
- A proprietary algorithm converts responses into a green, yellow, or red recommendation. In the applicant sample, 48% of applicants are green, 32% are yellow, and 20% are red.
- Testing is introduced gradually across locations. The paper defines test adoption at the location-month level as the first month in which the modal hire at that location has a test score, and all later months.
- HR managers receive the score and are encouraged to factor it into hiring, but they can still hire their preferred candidate.

Interpretation: The setting creates a natural comparison between pre-testing hiring and post-testing hiring, and it also creates variation in how closely managers follow test recommendations after the test becomes available.

3.2 Applicant and worker data

- The paper observes nearly 266,000 hired workers. About two-thirds are hired before testing and one-third after testing.
- In the post-testing period, the authors observe about 403,000 applicants, 91,000 hires, 445 HR managers, and 3,698 applicant pools.

- An applicant pool is defined at the manager-location-month level and includes applicants who applied within the previous four months and had not yet been hired.
- The average pool contains about 260 applicants, and the average hire probability is 19%.
- Test scores are predictive of hiring: greens are more likely to be hired than yellows, and yellows are more likely to be hired than reds.

Interpretation: The applicant-pool data are crucial. Without applicants who were not hired, it would be hard to know whether a manager actually chose a lower-scoring worker over an available higher-scoring worker.

3.3 Outcome measures

- The main outcome is job duration. Duration is important because turnover is costly and workers in this setting spend a substantial part of their early tenure in paid training.
- The completed-spell distribution is very short. The median completed spell is 99 days, and one in six workers leave after only one month.
- About 21% of spells are censored because the worker is still employed when the data end. The paper uses censored-normal regressions when log duration is the outcome.
- The paper also studies milestone outcomes: whether a worker survives at least 3, 6, or 12 months.
- For a subset of six client firms, the authors observe output per hour, a direct productivity measure. This is used to check whether managers trade off tenure for productivity.

Interpretation: Tenure is not a perfect measure of worker quality, but it is economically meaningful in this setting because turnover is common, costly, and linked to training costs. The productivity subsample helps address the concern that managers might hire short-tenure workers who are more productive while employed.

3.4 Exception rates

- The paper defines an exception as hiring against the test recommendation.
- A yellow hire is an exception when a green applicant had applied and was not hired.
- A red hire is an exception when a green or yellow applicant had applied and was not hired.
- The raw count of exceptions is normalized by the maximum number of exceptions that could have occurred in the applicant pool, given the pool's score composition and number of hires.
- The exception rate ranges from 0, meaning no possible exceptions are made, to 1, meaning all possible exceptions are made.
- The average exception rate across applicant pools is about 22%, and exception rates vary substantially across pools, managers, and locations.

Interpretation: The normalization is important because some pools mechanically allow more exceptions than others. The measure is designed to capture managerial choices rather than just the score distribution of the applicant pool.

3.5 Manager and location aggregation

- Manager-level exception rates are constructed by averaging across pools handled by the same manager, weighting by the number of hires.
- Location-level exception rates are constructed similarly when the analysis needs pre-testing observations, because manager identifiers are not observed in the pre-testing period.
- Manager-level variation is used to compare managers working at the same location after testing begins.
- Location-level variation is used to ask whether the treatment effect of testing is weaker in locations that later make more exceptions.

Interpretation: Aggregating exception rates helps smooth idiosyncratic applicant-pool shocks and makes the measure closer to a manager or location style of using discretion.

4 Models and empirical specifications

4.1 Effect of introducing the test

The first empirical specification estimates the effect of test availability on worker duration:

$$\log(\text{Duration})_{ilt} = \alpha_0 + \alpha_1 \text{Testing}_{lt} + \delta_l + \gamma_t + \text{Position}'_i \beta + \varepsilon_{ilt}. \quad (1)$$

- Duration_{ilt} is the job duration of worker i hired at location l in month t .
- Testing_{lt} equals one once testing is adopted at location l .
- Location fixed effects control for time-invariant differences across locations.
- Hire year-month fixed effects control for cohort and macroeconomic conditions.
- Position-type fixed effects control for small differences in job function.
- Richer specifications add client-firm-by-year fixed effects, local unemployment controls, and location-specific time trends.

Interpretation: The coefficient α_1 measures whether workers hired after the test is available stay longer than workers hired before the test, comparing within locations and controlling for time effects and job type.

4.2 Model of discretion

The paper presents a model in which worker quality for the firm is a_i , the test reveals a discrete type, and the HR manager observes additional private information but may also have bias or mistaken beliefs. The manager's utility from hiring worker i is

$$U_i = (1 - k)a_i + kb_i, \quad (2)$$

where b_i is an idiosyncratic manager-valued attribute and k measures the degree of bias or misalignment.

The manager also observes a noisy signal of firm-relevant worker quality:

$$s_i = a_i + \epsilon_i. \quad (3)$$

- If $k = 0$, the manager is aligned with the firm and cares only about firm-relevant quality.
- If k is large, the manager places more weight on attributes that may not maximize the firm's payoff.
- More precise private information can make discretion valuable.
- More bias or mistaken judgment can make discretion harmful.

Interpretation: Exceptions can come from two very different sources. They may reflect superior private information, or they may reflect bias and errors. The empirical question is which force dominates.

4.3 Exception-rate formula

For a manager m hiring at location l in month t , the paper defines

$$ExceptionRate_{m lt} = \frac{N_y^h N_g^{nh} + N_r^h (N_g^{nh} + N_y^{nh})}{\text{Maximum number of exceptions}}. \quad (4)$$

- N_y^h is the number of yellow applicants hired.
- N_g^{nh} is the number of green applicants not hired.
- N_r^h is the number of red applicants hired.
- N_y^{nh} is the number of yellow applicants not hired.
- The denominator adjusts for the maximum number of order violations possible in that pool.

Interpretation: The numerator counts lower-score hires made while higher-score applicants are left unhired. The denominator makes the measure comparable across applicant pools with different size, composition, and hiring needs.

4.4 Post-testing exception-outcome regression

The main post-testing discretion regression is

$$\log(Duration)_{imlt} = a_0 + a_1 ExceptionRate_m + \delta_l + \gamma_t + Position'_i \beta + \varepsilon_{imlt}. \quad (5)$$

- The sample is restricted to workers hired after testing is available.
- $ExceptionRate_m$ is the manager-level exception rate.
- Location fixed effects compare managers within the same location.
- Additional specifications add client-year effects, unemployment controls, location trends, and detailed applicant-pool controls.
- A negative a_1 means that workers hired by higher-exception managers stay for less time.

Interpretation: If exceptions primarily reflect useful private information, high-exception managers should have better outcomes. The paper finds the opposite, which points toward bias or mistakes.

4.5 Testing effects by exception rate

The paper also estimates whether the benefit of testing is weaker in locations that make more exceptions:

$$\log(\text{Duration})_{imlt} = b_0 + b_1 \text{Testing}_{lt} \times \text{ExceptionRate}_l + b_2 \text{Testing}_{lt} + \delta_l + \gamma_t + \text{Position}_i' \beta + \varepsilon_{imlt}. \quad (6)$$

- This specification uses both pre-testing and post-testing observations.
- Because manager identifiers are not observed before testing, the interaction uses a location-level exception rate.
- A negative b_1 means that testing improves outcomes less in high-exception locations.

Interpretation: If managers often ignore valuable test information, the treatment effect of testing should be attenuated where exception rates are high. The estimates show exactly this pattern.

4.6 Robustness and alternative explanations

- The paper compares exception hires with passed-over higher-scoring applicants who are hired later. If exceptions were based on superior private information, exception hires should outperform passed-over applicants.
- Instead, passed-over greens and yellows tend to stay longer than lower-scoring exception hires.
- The paper considers “false exceptions” that arise because job offers are unobserved. For example, a manager might appear to pass over a green applicant when in fact the green applicant rejected an offer.
- The results are robust to restrictions that reduce this concern, such as pools with enough green applicants to fill all hires.
- The paper studies output per hour to test whether managers choose workers with lower tenure but higher productivity. It finds no evidence of a positive productivity payoff from exceptions.

Interpretation: The robustness checks directly address the two main alternative stories: managers may be correctly using private information, or the exception rate may be mechanically generated by applicant yield rates. Neither explanation appears to drive the results.

5 Identification and empirical design

5.1 What the testing-adoption design identifies

- The test-adoption analysis exploits the staggered rollout of testing across locations.
- The key comparison is within a location before versus after testing, net of common time effects.
- Location fixed effects absorb permanent differences in turnover across locations.
- Year-month fixed effects absorb aggregate changes in the labor market and cohort composition.
- Client-year fixed effects and location trends reduce concerns that testing is introduced alongside other firm policies or location-specific trajectories.

Interpretation: The result that testing improves duration is credible because it uses timing variation across locations, but it should still be read as an observational design rather than a randomized experiment.

5.2 What the exception-rate design identifies

- The exception-rate analysis is not an experiment that randomly grants discretion to some managers and not others.
- Instead, it uses naturally occurring variation in how often managers appear to hire against the test.
- The key model-based logic is comparative: if discretion reflects valuable private information, high-exception managers should produce better hires; if discretion reflects bias or mistakes, high-exception managers should produce worse hires.
- The strongest post-testing specifications compare managers at the same location and control for observable applicant-pool composition.

Interpretation: The negative relationship between exceptions and tenure is informative because it goes in the opposite direction from the “manager has better information” hypothesis.

5.3 Applicant-pool and false-exception concerns

- A manager may appear to make exceptions simply because high-scoring applicants reject offers, but the data do not observe offers.
- A manager may face an unusually weak group of green applicants, making it optimal to hire a yellow or red applicant.
- The paper addresses these concerns by controlling for pool size, hire rate, share green, share yellow, location effects, local labor market conditions, and location trends.
- Aggregating exception rates to the manager level also reduces noise from one-off applicant-pool shocks.
- The passed-over applicant tests provide a direct check: higher-score workers who were initially passed over and hired later outperform the lower-score exception workers.

Interpretation: These checks do not make exceptions randomized, but they make it less plausible that the main result is entirely driven by applicant pools or unobserved offers.

5.4 Outcome interpretation

- Tenure is economically important because turnover is costly in this setting and training is paid.
- Tenure may still be incomplete if managers care about other dimensions of quality.
- The output-per-hour analysis is therefore important: it asks whether exceptions buy higher short-run productivity at the cost of lower retention.
- The paper finds no evidence that exceptions improve output per hour.

Interpretation: The evidence supports the view that exceptions worsen overall hiring quality, rather than merely changing the type of quality managers pursue.

5.5 External validity and limitations

- The job is low-skilled, standardized, and individual rather than team-based.
- HR managers in this setting are generally recruiters and are unlikely to supervise workers after hiring.
- The value of manager private information may be larger in high-skilled jobs, team-production settings, or jobs with complex match-specific requirements.
- The test is proprietary, so the paper cannot reveal all details about the algorithm or job.
- The results show that discretion is costly in this setting, not that human judgment is always useless in hiring.

Interpretation: The paper is most convincing as evidence about standardized service jobs and decision aids. It also provides a framework that other firms can apply to evaluate discretion in their own hiring processes.

6 Tables and figures

6.1 Table I and Figure I: Sample structure and job-duration distribution

Read:

- Table I shows that the authors observe 265,648 hired workers: 174,329 before testing and 91,319 after testing.
- The post-testing data contain 403,006 applicants, 445 HR managers, and 3,698 applicant pools.
- The applicant-pool panel shows that 48% of applicants are green, 32% are yellow, and 20% are red.
- Hire probabilities rise with test score: greens are hired more often than yellows, and yellows more often than reds.
- Figure I shows that completed job spells are very short and highly right-skewed. Many workers leave quickly, while a smaller number stay for long periods.

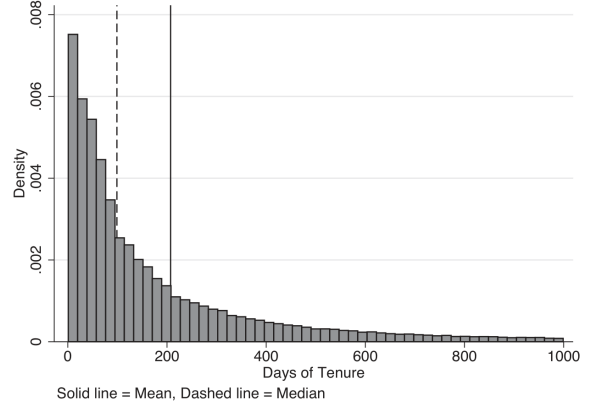
Economic message: The setting is one where worker retention is a central outcome. Because turnover is frequent and training is costly, a test that increases duration can have large economic value.

TABLE I
SUMMARY STATISTICS

	All	Pre-testing	Post-testing	
Sample coverage				
# Locations	127	113	97	
# Hired workers	265,648	174,329	91,319	
# Applicants			403,006	
# HR managers			445	
# Pools			3,698	
# Applicants/pool			260	
	Pre-testing	Post-testing	Green	Yellow
Worker characteristics mean (std. dev.)				
Duration of completed spell (days) ($N = 209,808$)	252 (323)	116 (138)	122 (143)	110 (121)
Duration of censored spell (days) ($N = 55,840$)	807 (510)	252 (245)	265 (252)	235 (232)
Share censored	0.19 (0.39)	0.25 (0.43)	0.24 (0.43)	0.26 (0.44)
Output per hour ($N = 62,427$)	8.35 (4.66)	8.44 (5.16)	8.39 (5.01)	9.16 (6.08)
		Post-testing	Green	Yellow
Applicant pool characteristics				
Share applicants			0.48	0.32
Hire probability		0.19	0.23	0.18

Notes. Post-testing is defined at the location-month level as the first month in which 50% of hires had test scores, and all months thereafter. An applicant pool is defined at the manager-location-month level and includes all applicants that had applied within four months of the current month and were not yet hired. Number of applicants reflects the total number of applicants across all pools. Applicant pool characteristics are unweighted averages across pools, and are calculated using individuals with test scores in the post-testing sample.

(a) Table I: Summary statistics



(b) Figure I: Distribution of completed job spells

6.2 Table II and Figure II: The effect of introducing testing

Read:

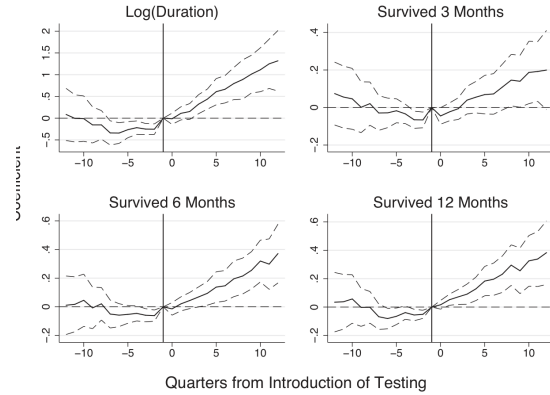
- Table II reports censored-normal regressions of log duration on test availability.
- The post-testing coefficient is positive and significant across all specifications.
- With full controls, the coefficient is 0.233, implying that workers hired after testing stay just over 25% longer.
- Figure II presents event-study evidence for log duration and 3-, 6-, and 12-month survival.
- The event-study patterns generally rise after testing is adopted, especially for longer retention milestones.

Economic message: Job testing contains useful information about applicant quality. Even though managers retain discretion, the introduction of the test improves retention relative to the firms' prior hiring practices.

TABLE II IMPACT OF JOB TESTING ON JOB DURATIONS				
	Dependent variable: log(Duration)			
	(1)	(2)	(3)	(4)
Post-testing	0.368*** (0.120)	0.244** (0.113)	0.248*** (0.0754)	0.233*** (0.0637)
N	265,648	265,648	265,648	265,648
Year-month FEs	X	X	X	X
Location FEs	X	X	X	X
Position-type FEs	X	X	X	X
Client firm $\times$ year FEs		X	X	X
Local unemployment controls			X	X
Location time trends				X

Notes. We regress log durations on an indicator for testing availability (this equals 1 in the first month in which the modal hire at a location was tested, and in all months thereafter for that location) and the controls indicated. We use censored-normal regressions with individual-specific truncation points (using "onreg" in Stata) to account for the fact that 21% of hired workers had not yet left their job at the end of our data collection. Standard errors are in parentheses and are clustered at the location level. FE = fixed effects. *** $p < .01$, ** $p < .05$, * $p < .1$.

(a) Table II: Impact of job testing on job durations



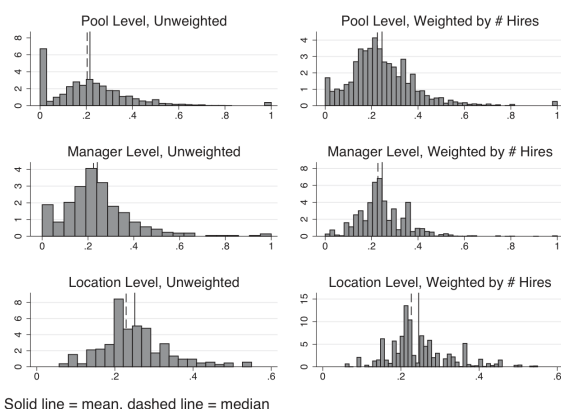
(b) Figure II: Event study of duration outcomes

6.3 Figure III and Figure IV: Exception-rate variation and manager outcomes

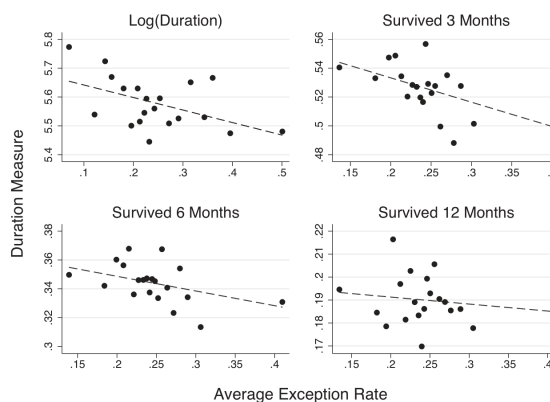
Read:

- Figure III shows that exceptions are common. The average applicant-pool exception rate is about 22%, and there is meaningful variation across pools, managers, and locations.
- The manager-level plots show that some managers consistently make more exceptions than others.
- Figure IV bins managers by exception rate and plots worker-duration outcomes.
- For log duration and the 3-, 6-, and 12-month survival measures, the fitted relationships are negative: higher exception rates are associated with worse worker retention.

Economic message: Discretion is not rare noise in this setting. It is a systematic managerial style, and the managers who use more discretion against the test tend to hire workers who leave sooner.



(a) Figure III: Distributions of exception rates



(b) Figure IV: Manager-level exception rates and durations

6.4 Table III and Table IV: Regression evidence on exceptions

Read:

- Table III estimates post-testing regressions of log duration on the manager-level exception rate.
- The exception-rate coefficient is negative and statistically significant in every specification.
- In the full specification with applicant-pool controls, a one-standard-deviation increase in manager exception rate is associated with a 0.0557 log-point reduction in duration.
- Table IV asks whether the impact of testing is weaker in locations that make more exceptions.
- The interaction between location exception rate and post-testing is negative and significant, around -0.14 to -0.15 across specifications.

Economic message: Table III shows that high-exception managers hire worse-retention workers after testing is available. Table IV shows that the benefits of testing are smaller where managers more often overrule the test.

TABLE III
EXCEPTION RATES AND POST-TESTING DURATION

	Dependent variable: log(Duration)				
	(1)	(2)	(3)	(4)	(5)
Manager exception rate	-0.0682** (0.0346)	-0.0658** (0.0321)	-0.0661** (0.0322)	-0.0607** (0.0292)	-0.0557** (0.0283)
<i>N</i>	91,319	91,319	91,319	91,319	91,319
Year-month FEs	X	X	X	X	X
Location FEs	X	X	X	X	X
Position-type FEs	X	X	X	X	X
Client firm $\times$ year FEs		X	X	X	X
Local unemployment controls			X	X	X
Location time trends				X	X
Applicant pool controls					X

Notes. This table reports censored normal regressions of manager-level exception rates and tenure outcomes restricted to the post-testing sample. The exception rate is defined as the number of times a yellow is hired above a green plus the number of times a red is hired above a green or yellow, divided by the maximum exceptions possible in that applicant pool. It is then aggregated to the manager level and standardized to be mean 0 and standard deviation 1. Applicant pool controls include fixed effects for deciles of each of the following variables: number of applicants, hire rate, share of applicants that are green, and share that are

(a) Table III: Exception rates and post-testing duration

TABLE IV
THE IMPACT OF TESTING BY EXCEPTION RATE

	Dependent variable: log(Duration)			
	(1)	(2)	(3)	(4)
Post-testing	0.366*** (0.119)	0.234** (0.107)	0.242*** (0.0696)	0.234*** (0.0589)
Location exception	-0.142** (0.0652)	-0.150** (0.0679)	-0.152** (0.0655)	-0.142** (0.0573)
Rate $\times$ post-testing				
<i>N</i>	265,648	265,648	265,648	265,648
Year-month FEs	X	X	X	X
Location FEs	X	X	X	X
Position-type FEs	X	X	X	X
Client firm $\times$ year FEs		X	X	X
Local unemployment controls			X	X
Location time trends				X

Notes. This table reports censored normal regressions of the differential impact of testing adoption, by location-level exception rate. We use the same sample as defined by the notes to Table II. The exception rate is defined as the number of times a yellow is hired above a green plus the number of times a red is hired above

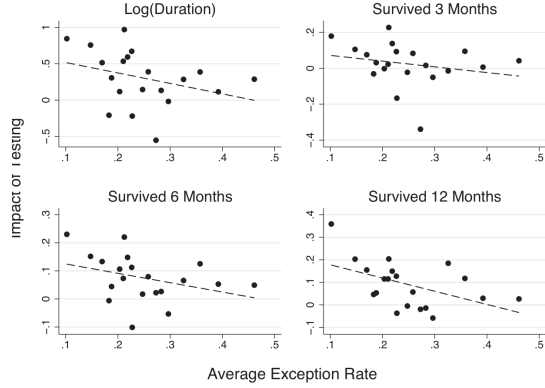
(b) Table IV: The impact of testing by exception rate

6.5 Figure V and Table V: Location-level attenuation and passed-over applicants

Read:

- Figure V plots the effect of testing across location-level exception-rate bins.
- The impact of testing is lower in higher-exception locations for log duration and for the 3-, 6-, and 12-month survival outcomes.
- Table V compares lower-score exception hires with higher-score applicants who were available at the same time but hired later.
- Passed-over greens do not perform worse than yellow exceptions; for red exceptions, passed-over greens and yellows perform substantially better.
- In the red-exception comparison with applicant-pool fixed effects, passed-over greens stay about 14% longer and passed-over yellows about 12% longer than the red exception hires.

Economic message: The passed-over applicant evidence is especially important. It directly contradicts the idea that managers are usually making exceptions because they know that the lower-score applicant is better.



(a) Figure V: Location-level exception rates and testing effects

TABLE V
TENURE OF EXCEPTIONS VERSUS PASSED OVER APPLICANTS

	Dependent variable: log(Duration)	
	(1)	(2)
Quality of yellow exceptions versus passed over greens		
Passed over greens	0.0402* (0.0220)	0.0449 (0.0357)
N	53,166	53,166
Quality of red exceptions versus passed over greens and yellows		
Passed over greens	0.159*** (0.0543)	0.143** (0.0634)
Passed over yellows	0.143*** (0.0546)	0.121** (0.0597)
N	25,782	25,782
Base controls	X	X
Comparison pool FEs		X

Notes. Regressions are restricted to the post-testing sample, adjust for censoring, and standard errors are clustered at the location level. The top (bottom) panel compares yellow (red) exceptions—the omitted category—to passed over greens (and yellows) who were available at the same time but hired in a later month. Observations are restricted to pools with at least one exception and one passed over worker, and are further restricted to locations and pools with at least 10 and 5 observations, respectively. Base controls are location, hire month, and position-type fixed effects. Comparison pool fixed effects are defined by the manager–location–month for the applicant pool in which candidates were considered together. *** $p < .01$. ** $p < .05$. * $p < .1$.

(b) Table V: Tenure of exceptions versus passed-over applicants

6.6 Table VI: Productivity and the tenure-productivity trade-off

Read:

- Table VI uses output per hour as the dependent variable for the subset of firms with productivity data.
- The post-testing sample regressions show no significant positive relation between manager exception rates and output per hour.
- The testing-by-exception-rate specifications also show no robust positive productivity effect for high-exception locations.
- The coefficients are noisy, but the table does not support the hypothesis that managers sacrifice tenure to hire more productive workers.

Economic message: The exception penalty is not offset by higher observed productivity. This makes it more likely that exceptions reflect poor hiring judgment rather than an optimal trade-off across worker-quality dimensions.

TABLE VI
TESTING, EXCEPTION RATES, AND OUTPUT PER HOUR

	Dependent variable: output per hour			
	(1)	(2)	(3)	(4)
Post-testing	Post-testing sample		Introduction of testing	
			0.343	0.156
			(0.366)	(0.327)
Exception rate $\times$ post-testing	-0.0659 (0.134)	-0.111 (0.0953)	0.137 (0.153)	-0.137 (0.217)
N	28,858	28,858	62,421	62,421
Base controls	X	X	X	X
Full controls		X		X

Notes. See notes to Tables III and IV. The dependent variable in this case is output per hour and regressions are estimated with OLS. In columns (1) and (2), we examine the post-testing correlation between the manager-level exception rate and output per hour. In columns (3) and (4), we examine the differential impact of testing as a function of location-level exception rates. Base controls include location, hire month, and position fixed effects, as well as controls for the number of tasks in a day that are used to measure a worker's output per hour. Full controls add client-by-year effects, local unemployment rates, and location-specific time trends. For the post-testing sample regressions (columns (1) and (2)), full controls also include applicant pool controls. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table VI: Testing, exception rates, and output per hour

7 Short summary

- The paper studies 15 firms that introduce a job test for a standardized low-skilled service-sector job.
- The test assigns applicants green, yellow, or red scores.
- Managers see the test score but retain discretion over whom to hire.
- Introducing the test increases worker tenure substantially.
- The authors define exceptions as lower-score hires made while higher-score applicants are left unhired.
- Exceptions are common and vary systematically across managers and locations.
- Higher manager exception rates predict shorter job durations.
- Locations with more exceptions benefit less from testing.
- Passed-over higher-score applicants later outperform lower-score exception hires.
- Exceptions do not appear to improve output per hour.
- The evidence suggests that managers often overrule test recommendations because of bias or mistakes, not because of superior private information.

8 Conclusion

8.1 Core conclusion

Hoffman, Kahn, and Li show that job testing can improve hiring outcomes, but that the organizational use of the test matters. In their low-skilled service-sector setting, the introduction of testing increases worker tenure. However, managers who frequently overrule the test by hiring lower-score applicants over available higher-score applicants systematically hire workers who leave sooner.

8.2 Economic intuition

The economic intuition is a trade-off between information and agency. Managers may have private information from interviews and resumes that the test does not capture, so some discretion could be valuable. But managers may also have biases, misaligned preferences, or overconfidence in their own judgment. If the private-information channel dominates, exceptions should improve outcomes. If bias and mistakes dominate, exceptions should worsen outcomes. The paper finds the latter pattern.

8.3 Incremental contribution revisited

The paper advances the literature by studying not only whether hiring tests predict outcomes, but also whether managers add value when they can deviate from those tests. It creates a practical empirical measure of discretion, links it to a simple model of information and bias, and tests the predictions using applicant-level and worker-level data. This makes the paper an important empirical contribution to workforce analytics, personnel economics, and organizational decision rights.

8.4 Implications for firms

The results imply that firms should not merely buy screening technology and leave the rest of the hiring process unchanged. They should monitor how managers use algorithmic recommendations, identify managers with high exception rates, provide feedback on hiring outcomes, and consider limits on discretion when exceptions are systematically associated with worse performance. A test may be especially valuable because it gives the firm a verifiable signal that reduces information asymmetry between headquarters and local hiring managers.

8.5 Limitations

The results come from standardized low-skilled service jobs, so the value of discretion may differ in higher-skilled or team-based settings. The data do not observe job offers, which creates the possibility of false exceptions, although the paper addresses this concern with several checks. The design is observational, not a randomized experiment in discretion. Finally, the test is proprietary, so the paper cannot fully reveal the algorithm or job details.

8.6 Takeaway

The enduring takeaway is that human discretion is not automatically complementary to algorithmic screening. In this setting, managers who rely less on the test make worse hires on average. The paper therefore turns a broad debate about human judgment versus algorithms into a concrete organizational lesson: decision aids are most valuable when firms also manage how employees use them.